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W10_D4
CSPT25

Simulazione fase di raccolta informazioni pt. 2

Traccia

https://www.yeahhub.com/15-most-useful-host-scanning-commands-kalilinux/

Utilizzare alcuni di questi strumenti per raccogliere informazioni sulla macchina metasploitable e produrre un report.

Nel report indicare sopra l'esecuzione degli strumenti e nella parte finale un riepilogo delle informazioni trovate

Facoltativo:

Utilizzare tutti i tool proposti ed approfondire lo studio dei metodi di evasione firewall con Nmap:

- https://nmap.org/book/firewall-subversion.html
- https://nmap.org/book/man-bypass-firewalls-ids.html

Per questa simulazione utilizzeremo due macchine virtuali (Metasploitable2 & Kali Linux) impostate nella stessa rete.

Macchina Target: **Metaploitable2**
IP: **192.168.50.101**
Abiente: **rete laboratorio virtuale in statica.**

Eseguiremo i seguenti comandi per scannerizzare le porte del target:

nmap -sn -PE: verifichiamo se l'host è attivo.

```
kali@kali: ~
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sn -PE 192.168.50.101
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:23 -0500
Nmap scan report for 192.168.50.101
Host is up (0.0015s latency).
MAC Address: 8A:F3:35:BF:62:D6 (Unknown)
Nmap done: 1 IP address (1 host up) scanned in 4.10 seconds
```

netdiscover -r: per trovare host attivi in LAN

```
kali@kali: ~
Session  Actions  Edit  View  Help
 Currently scanning: Finished!   |   Screen View: Unique Hosts

 1 Captured ARP Req/Rep packets, from 1 hosts.   Total size: 60
 _____________________________________________________________________________
   IP            At MAC Address     Count     Len  MAC Vendor / Hostname
 -----------------------------------------------------------------------------
 192.168.50.101  8a:f3:35:bf:62:d6      1      60  Unknown vendor
```

nmap *IP TARGET* —top-ports 10 —open: per scannerizzare le 1p porte più comuni aperte.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap 192.168.50.101 --top-ports 10 --open
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:27 -0500
Nmap scan report for 192.168.50.101
Host is up (0.0078s latency).
Not shown: 3 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
80/tcp   open  http
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
MAC Address: 8A:F3:35:BF:62:D6 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 4.14 seconds
```

nmap -sV *IP TARGET*: per identificare la versione dei servizi.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV 192.168.50.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:28 -0500
Nmap scan report for 192.168.50.101
Host is up (0.0010s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login?
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 8A:F3:35:BF:62:D6 (Unknown)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; C
PE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
.
Nmap done: 1 IP address (1 host up) scanned in 56.76 seconds
```

nmap -sS -sV -T4: per eseguire uno scan completo.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sS -sV -T4 192.168.50.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:30 -0500
Nmap scan report for 192.168.50.101
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login?
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 8A:F3:35:BF:62:D6 (Unknown)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; C
PE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
.
Nmap done: 1 IP address (1 host up) scanned in 56.80 seconds
```

nmap -p-: scansioniamo le porte da 1 a 65535.

```
└─$ sudo nmap -p- 192.168.50.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:34 -0500
Nmap scan report for 192.168.50.101
Host is up (0.0010s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
38723/tcp open  unknown
45468/tcp open  unknown
47996/tcp open  unknown
59724/tcp open  unknown
MAC Address: 8A:F3:35:BF:62:D6 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 7.39 seconds
```

nc -nvz *IP TARGET* 1-1024: Scansione Netcat alle porte comuni da 1-1024.

```
┌──(kali㉿kali)-[~]
└─$ nc -nvz 192.168.50.101 1-1024
(UNKNOWN) [192.168.50.101] 514 (shell) open
(UNKNOWN) [192.168.50.101] 513 (login) open
(UNKNOWN) [192.168.50.101] 512 (exec) open
(UNKNOWN) [192.168.50.101] 445 (microsoft-ds) open
(UNKNOWN) [192.168.50.101] 139 (netbios-ssn) open
(UNKNOWN) [192.168.50.101] 111 (sunrpc) open
(UNKNOWN) [192.168.50.101] 80 (http) open
(UNKNOWN) [192.168.50.101] 53 (domain) open
(UNKNOWN) [192.168.50.101] 25 (smtp) open
(UNKNOWN) [192.168.50.101] 23 (telnet) open
(UNKNOWN) [192.168.50.101] 22 (ssh) open
(UNKNOWN) [192.168.50.101] 21 (ftp) open
```

Nc -nv *IP TARGET* 22: Scansione Netcat alla porta specifica.

```
┌──(kali㉿kali)-[~]
└─$ nc -nv 192.168.50.101 22
(UNKNOWN) [192.168.50.101] 22 (ssh) open
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
```

nmap -f —mtu=512: Scan frammentando i pacchetti ed impostando la dimensione per bypassare firewall semplici.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -f --mtu=512 192.168.50.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:41 -0500
Nmap scan report for 192.168.50.101
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 8A:F3:35:BF:62:D6 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 4.26 seconds
```

Masscan *IP TARGET* -p80 —banners: per eseguire uno scan di rete molto veloce.

```
┌──(kali㉿kali)-[~]
└─$ sudo masscan 192.168.50.101 -p80 --banners
[+] resolving router 192.168.50.100 with ARP (may take some time)...
[-] FAIL: ARP timed-out resolving MAC address for router eth0: "0.0.0.0"
    [hint] try "--router ip 192.0.2.1" to specify different router
    [hint] try "--router-mac 66-55-44-33-22-11" instead to bypass ARP
    [hint] try "--interface eth0" to change interface
```

Osservazioni:

Grazie a tutti questi comandi per scannerizzare le porte del target si può osservare che il target metasploitable2 espone numerose porte di servizi (FTP, Telnet, HTTP, MySQL).